although this is seldom the most efficient method to compute it (this series converges for all matrices $\boldsymbol{A}$, and so in fact the exponential expression for the solution of the linear system is well defined even in the absence of a full set of eigenvectors). Typically one favours diagonalization of the matrix $\boldsymbol{A}$ in order to exponentiate (for more details see for instance [52, 273, 275]).

Noting that the origin of time is arbitrary in an autonomous system such as this one, we may think of the exponential as a representation of the flow map,

$$\mathscr{F}_t(\boldsymbol{z}) \equiv e^{\boldsymbol{A}t}\boldsymbol{z}.$$

1.5.2 Integrable Examples

Given a differential equation system, the first instinct is to try to solve the equations. Being able to do this relies on having enough constants of motion available to "pin down" the solution. Systems whose solutions are constrained in this way are examples of *integrable systems*.

As mentioned previously, for the N-body system, energy and total momentum are constants of motion—they do not change even as the bodies of the system move along their natural paths (defined by the equations of motion). Another term for constant of motion is *first integral*. In general, if we have a dynamical system $\dot{\boldsymbol{z}} = \boldsymbol{f}(\boldsymbol{z})$, a *first integral* is a smooth function $I(\boldsymbol{z})$ which is constant along solutions, for all values of the initial condition. Let $\boldsymbol{z}(t) : t \in \mathbb{R}$ be a solution, then

$$I(\boldsymbol{z}(t)) \equiv I(\boldsymbol{z}(0)) \Rightarrow 0 \equiv \frac{d}{dt} I(\boldsymbol{z}(t)) = \nabla I(\boldsymbol{z}(t)) \cdot \dot{\boldsymbol{z}}(t) = \nabla I(\boldsymbol{z}(t)) \cdot \boldsymbol{f}(\boldsymbol{z}(t)).$$

Since this should hold everywhere, the condition for I to be a first integral is that $\nabla I \cdot \boldsymbol{f} \equiv 0$.

The existence of a first integral of a two-dimensional system of ordinary differential equations $\dot{x} = g(x, y), \dot{y} = h(x, y)$ allows the description of all the solutions of the equation in terms of the level curves of a scalar-valued function I: $I(x, y) \equiv I_0$.

In principle such an equation can be solved (locally at least) for y as a function of x due to the implicit function theorem. Hence one may write $y = \psi(x)$. Reinsert this into the first differential equation to get a reduced equation in just one dependent variable:

$$\frac{\mathrm{d}x}{\mathrm{d}t} = g(x, \psi(x)).$$

Such an ordinary differential equation is said to be *separable*, and theoretically can be solved, given an initial condition, for x as a function of t.

Example 1.6 The single degree of freedom model of Example 1.4 has the energy as a first integral. The system is therefore integrable.